

1 Solutions of Problems in Exercise Session 2

Exercise 1.1 You observe the following four episodes (state, reward, next state, next reward, ...): $(s_1, 0, s_3, 1, s_4, 0)$, $(s_1, 0, s_3, 0, s_4, 0)$, $(s_1, 0, s_3, 1, s_4, 0)$, $(s_2, 0, s_3, 0, s_4, 0)$. Assume zero initialization and $\alpha_i = 1/i$ for TD, where i is the episode index ($i = 1, \dots, 4$). Determine:

- (a) the values estimated by Monte Carlo,
- (b) the values estimated by TD,
- (c) the true state values V ,
- (d) an intuitive explanation for the difference in the estimate of $V(s_2)$ between the two methods,
- (e) what happens if you keep iterating both algorithms over these four episodes.

Solution. (a) We begin by calculating the MC estimate V_{MC} . The initial estimate is $V^{(0)}(s) = 0$ for all states s . In the first episode, the samples are $(s_1, 0, s_3, 1, s_4, 0)$, implying that the discounted returns are $G_{t=1} = 0 + \gamma + 0 = \gamma$, $G_{t=2} = 1 + 0 = 1$, and $G_{t=3} = 0$. Thus, we perform two MC updates after the first episode $i = 1$:

$$V^{(1)}(s_1) = V^{(0)}(s_1) + \frac{1}{C_1(s_1)}(G_{t=1} - V^{(0)}(s_1)) = G_{t=1} = \gamma$$

$$V^{(1)}(s_3) = V^{(0)}(s_3) + \frac{1}{C_1(s_3)}(G_{t=2} - V^{(0)}(s_3)) = G_{t=2} = 1$$

Note that, when we do not explicitly write expressions for $V^{(1)}(s)$ this means that $V^{(1)}(s) = V^{(0)}(s)$ on states s we do not mention (in this case $V^{(1)}(s_2) = V^{(0)}(s_2) = 0$, $V^{(1)}(s_4) = V^{(0)}(s_4) = 0$). We repeat this procedure for the remaining three episodes. In the second episode ($i = 2$), where all returns are zero, only non-zero estimates are updated:

$$V^{(2)}(s_1) = V^{(1)}(s_1) + \frac{1}{C_2(s_1)}(G_{t=1} - V^{(1)}(s_1)) = \left(1 - \frac{1}{2}\right) V^{(1)}(s_1) = \frac{1}{2}\gamma$$

$$V^{(2)}(s_3) = V^{(1)}(s_3) + \frac{1}{C_2(s_3)}(G_{t=2} - V^{(1)}(s_3)) = \left(1 - \frac{1}{2}\right) V^{(1)}(s_3) = \frac{1}{2}$$

In the third episode, the samples are the same as in the first episode, so $G_{t=1} = \gamma$ and $G_{t=2} = 1$, updating s_1 and s_3 again:

$$V^{(3)}(s_1) = V^{(2)}(s_1) + \frac{1}{C_3(s_1)}(G_{t=1} - V^{(2)}(s_1)) = \frac{2}{3}\gamma$$

$$V^{(3)}(s_3) = V^{(2)}(s_3) + \frac{1}{C_3(s_3)}(G_{t=2} - V^{(2)}(s_3)) = \frac{2}{3}$$

Finally, in the last episode, with no non-zero returns, only the non-zero estimate of s_3 is updated:

$$V^{(4)}(s_3) = V^{(3)}(s_3) + \frac{1}{C_4(s_3)}(G_{t=2} - V^{(3)}(s_3)) = \frac{1}{2}$$

Thus, the final MC estimate is: $V_{MC} = [\frac{2}{3}\gamma, 0, \frac{1}{2}, 0]$.

(b) Next, we calculate the TD estimate V_{TD} . Starting with zero-initialized estimates, we update then when a non-zero reward is observed. In the second step of the first episode, we observe reward of 1 in s_3 , leading to an update:

$$V^{(2)}(s_3) = V^{(1)}(s_3) + \alpha(r_2 + \gamma V^{(1)}(s_4) - V^{(1)}(s_3)) = \frac{1}{i}r_2 = 1$$

Next update happens after state s_1 in the second episode, since state s_1 is followed by state s_3 with non-zero value estimate:

$$\begin{aligned} V^{(4)}(s_1) &= V^{(3)}(s_1) + \alpha(r_4 + \gamma V^{(3)}(s_3) - V^{(3)}(s_1)) = \frac{1}{i} \gamma V^{(3)}(s_3) = \frac{1}{2} \gamma \\ V^{(5)}(s_3) &= V^{(4)}(s_3) + \alpha(r_5 + \gamma V^{(4)}(s_4) - V^{(4)}(s_3)) = \left(1 - \frac{1}{i}\right) V^{(4)}(s_3) = \frac{1}{2} \end{aligned}$$

During the third episode, we update $V^{(7)}(s_1) = \frac{1}{2} \gamma$, $V^{(8)}(s_3) = \frac{2}{3}$. Now, note that in the last episode we update the value estimate of state s_2 since it is followed by state s_3 :

$$\begin{aligned} V^{(10)}(s_2) &= V^{(9)}(s_2) + \alpha(r_{10} + \gamma V^{(9)}(s_3) - V^{(9)}(s_2)) = \frac{1}{i} \gamma V^{(9)}(s_3) = \frac{1}{6} \gamma \\ V^{(11)}(s_3) &= V^{(10)}(s_3) + \alpha(r_{11} + \gamma V^{(10)}(s_4) - V^{(10)}(s_3)) = \frac{1}{2} \end{aligned}$$

yielding final estimate: $V_{\text{TD}} = [\frac{1}{2} \gamma, \frac{1}{6} \gamma, \frac{1}{2}, 0]$.

- (c) The true values of the states are $V(s_1) = \frac{1}{2} \gamma$, $V(s_2) = \frac{1}{2} \gamma$, $V(s_3) = \frac{1}{2}$, $V(s_4) = 0$.
- (d) Note that $V_{\text{MC}}(s_2) = 0$, while $V_{\text{TD}}(s_2) = \frac{1}{6} \gamma$. This difference arises because TD methods use bootstrapping, while MC methods do not. MC methods rely only on observed returns, while TD also considers the estimated values of subsequent states. Since all observed rewards in the last episode (the only one where s_2 is visited) are zero, MC methods provide us with $V_{\text{MC}}(s_2) = 0$ as an estimate. On the other hand, TD methods take into consideration that the state s_2 is followed by state s_3 that has non-zero value estimate, and thus updates $V_{\text{TD}}(s_2)$ to a non-zero value.
- (e) The only thing that would change is that $V_{\text{TD}}(s_2)$ would converge to $V(s_2) = \frac{1}{2} \gamma$.

Exercise 1.2 Consider the MDP shown in Figure 1. Show that the variance of the TD targets for $V(s_2)$ decreases as more samples are collected, whereas the variance of the MC targets remains constant.

Solution. We begin with the Monte Carlo (MC) case. The only stochasticity in this MDP comes from the reward received after visiting s_3 , which is $r(s_3) \sim \text{Bern}(0.5)$. From s_2 , the return is therefore

$$G_{s_2} = \gamma r(s_3),$$

so G_{s_2} is a $\gamma \text{Bern}(0.5)$ random variable. Hence,

$$\mathbb{E}[G_{s_2}] = 0.5\gamma, \quad \text{Var}[G_{s_2}] = \gamma^2 \cdot 0.5(1 - 0.5) = \frac{1}{4} \gamma^2.$$

Since the MC target for $V(s_2)$ is exactly this return G_{s_2} , its variance remains fixed at $\frac{1}{4} \gamma^2$, regardless of how many samples we collect. Averaging over more samples will reduce the variance of the *estimate* of $V(s_2)$, but each individual target has this constant variance.

Now, consider the Temporal-Difference (TD) case. The TD target for $V(s_2)$ is

$$r(s_2) + \gamma V(s_3) = \gamma V(s_3),$$

since $r(s_2) = 0$. Unlike in MC, this target depends on the current estimate of $V(s_3)$ rather than on a single sampled return. After k visits to s_2 , we expect roughly $10k$ visits to s_3 (since s_1 is visited nine times more often). Thus, $V(s_3)$ will be estimated as the empirical mean of approximately $10k$ samples of $r(s_3) \sim \text{Bern}(0.5)$, i.e.

$$V(s_3) \approx \frac{1}{10k} \sum_{i=1}^{10k} r_i, \quad \sum_{i=1}^{10k} r_i \sim \text{Bin}(10k, 0.5).$$

The variance of this average is

$$\text{Var}[V(s_3)] = \frac{0.5(1 - 0.5)}{10k} = \frac{1}{40k}.$$

Consequently, the TD target variance for $V(s_2)$ is

$$\text{Var}[\gamma V(s_3)] = \gamma^2 \cdot \frac{1}{40k},$$

which decreases proportionally to $1/k$ as more samples are collected.

Exercise 1.3 For the MDP shown in Figure 2, estimate how many steps each method requires to accurately approximate $V(s_2)$, assuming a learning rate of $\alpha = 1$.

Solution. The first update occurs only after the whole episode ends, i.e., when state s_{100} is reached. The MC method updates all states that have been observed in that episode. Depending on the initial state, either $V(s_1)$ or $V(s_2)$ is updated. Note that, since rewards are deterministic, all values are updated to their true values - except for states s_1 or s_2 . Since the probability of starting at state s_2 is 0.1, this means that, on average, we need 10 episodes to update s_2 to its true value using the MC method.

On the other hand, the TD method updates only $V(s_{99})$ after the first episode ($t = 1$), and all other values remain zero. In the next episode, $V(s_{98})$ is updated ($t = \gamma$), in the episode after that $V(s_{97})$, and so on. This means that we need exactly 97 episodes until state s_3 is updated. After that, we need to observe state s_2 to update it using estimate $V(s_3)$. Again, since the probability of starting in s_2 is 0.1, this would require, on average, 10 more episodes, for a total of 107 episodes. In this constructed example, we see that MC methods utilize the structure of the MDP much better, leading to 10 times fewer transitions needed for convergence.

Exercise 1.4 Show that the total updates made by the offline λ -return algorithm are equal to those made by the online TD(λ) algorithm (using averaging eligibility traces) - provided we accumulate the updates over the entire episode but apply them only once at the end ($T = \infty$). In other words, show that:

$$\sum_{t=0}^{\infty} (G_t^\lambda - V(s)) \mathbf{1}_{(s=s_t)} = \sum_{t=0}^{\infty} z_t(s) (r_t + \gamma V(s_{t+1}) - V(s_t))$$

Step 1: Express eligibility traces recursively. By definition of averaging traces:

$$z_t(s) = \mathbf{1}_{(s=s_t)} + \gamma \lambda z_{t-1}(s).$$

Starting with $z_0(s) = \mathbf{1}_{(s=s_0)}$ and unrolling this recursion gives:

$$z_t(s) = \sum_{k=0}^t (\gamma \lambda)^{t-k} \mathbf{1}_{(s=s_k)}.$$

Step 2: Express n -step returns in terms of TD errors. Recall definition of n -step return:

$$G_{t:t+n} = r_t + \gamma r_{t+1} + \dots + \gamma^{n-1} r_{t+n-1} + \gamma^n V(s_{t+n})$$

Next we can check that for TD errors $\delta_{t+\ell} = r_{t+\ell} + \gamma V(s_{t+\ell+1}) - V(s_{t+\ell})$ we have:

$$\begin{aligned}
V(s_t) + \sum_{\ell=0}^{n-1} \gamma^\ell \delta_{t+\ell} &= \underbrace{V(s_t) + \delta_t}_{r_t + \gamma V(s_{t+1})} + \gamma \delta_{t+1} + \cdots + \gamma^{n-1} \delta_{t+n-1} \\
&= r_t + \gamma \underbrace{(V(s_{t+1}) + \delta_{t+1})}_{r_{t+1} + \gamma V(s_{t+2})} + \cdots + \gamma^{n-1} \delta_{t+n-1} \\
&= \cdots \\
&= r_t + \gamma r_{t+1} + \cdots + \gamma^{n-1} r_{t+n-1} + \gamma^n V(s_{t+n}) \\
&= G_{t:t+n},
\end{aligned}$$

exactly as we wanted to show.

Step 3: Express the λ -return using TD errors. Using the definition of the λ -return and results from step 2:

$$\begin{aligned}
G_t^\lambda &= (1 - \lambda) \sum_{n=1}^{\infty} \lambda^{n-1} G_{t:t+n} = (1 - \lambda) \sum_{n=1}^{\infty} \lambda^{n-1} \left(V(s_t) + \sum_{\ell=0}^{n-1} \gamma^\ell \delta_{t+\ell} \right) \\
&= (1 - \lambda) V(s_t) \underbrace{\sum_{n=1}^{\infty} \lambda^{n-1}}_{1/(1-\lambda)} + (1 - \lambda) \sum_{n=1}^{\infty} \sum_{\ell=0}^{n-1} \lambda^{n-1} \gamma^\ell \delta_{t+\ell}
\end{aligned}$$

The double sum contains sums of the form $\lambda^{n-1}(\delta_t + \gamma \delta_{t+1} + \cdots + \gamma^{n-1} \delta_{t+n-1})$ for each n , and hence each TD error δ_{t+k} appears in the final sum multiplied by $\gamma^k \sum_{n=k+1}^{\infty} \lambda^{n-1} = (\gamma\lambda)^k / (1 - \lambda)$. Thus, finally we obtain:

$$G_t^\lambda - V(s_t) = \sum_{k=0}^{\infty} (\gamma\lambda)^k \delta_{t+k}.$$

Finally, we have all parts to show the final equation. First, consider the total offline update:

$$\sum_{t=0}^{\infty} (G_t^\lambda - V(s_t)) \mathbf{1}_{(s=s_t)} = \sum_{t=0}^{\infty} \mathbf{1}_{(s=s_t)} \sum_{k=0}^{\infty} (\gamma\lambda)^k \delta_{t+k}.$$

From first step we know that $z_t(s) = \sum_{k=0}^t (\gamma\lambda)^{t-k} \mathbf{1}_{(s=s_k)}$, and thus we also have:

$$\sum_{t=0}^{\infty} z_t(s) (r_t + \gamma V(s_{t+1}) - V(s_t)) = \sum_{t=0}^{\infty} z_t(s) \delta_t = \sum_{t=0}^{\infty} \sum_{k=0}^t (\gamma\lambda)^{t-k} \mathbf{1}_{(s=s_k)} \delta_t$$

Thus, the only thing left to show is that:

$$\sum_{t=0}^{\infty} \sum_{k=0}^{\infty} (\gamma\lambda)^k \mathbf{1}_{(s=s_t)} \delta_{t+k} = \sum_{t=0}^{\infty} \sum_{k=0}^t (\gamma\lambda)^{t-k} \mathbf{1}_{(s=s_k)} \delta_t,$$

but this is straightforward by comparing the terms next to $\mathbf{1}_{(s=s_t)}$ which are of form $\delta_t + (\gamma\lambda)\delta_{t+1} + (\gamma\lambda)^2\delta_{t+2} + \cdots$ in the both sums.